

Project Approval System (PAS)

Professional Documentation Report Draft

Student Name: [Insert full name]

Student ID: [Insert student ID]

Current Live Location: Colombo, Sri Lanka

Module: PUSL2020 Software Development Tools and Practices

This report documents the design, implementation, and quality assurance evidence for a premium web-based Project Approval System. Replace each screenshot placeholder with the actual screenshots captured from the running Windows laptop environment.

1. Executive Summary

The Project Approval System is implemented as an ASP.NET Core MVC application that helps match student research proposals with academic supervisors. It includes client-server interaction, geospatial display, MVC data rendering, SQLite persistence, database search, Regex validation, a premium dashboard, and an admin command center.

Premium additions include a dashboard, admin panel, proposal review flow, seeded supervisor registry, and a formal QA evidence checklist.

2. Architecture Overview

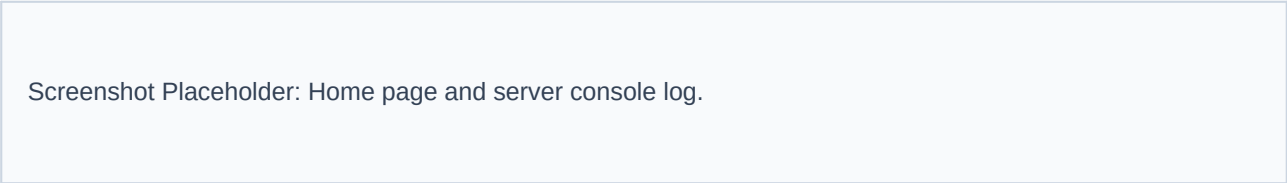
The solution follows an MVC architecture with separated Models, Views, Controllers, Services, and Data layers. Entity Framework Core with SQLite is used for lightweight persistence. Student identity details are centralized through a service and logged on each major request path.

Layer	Implementation
Models	Supervisor, ProposalSubmission, ProposalRecord, StudentIdentity, view models
Controllers	HomeController for public workflows, AdminController for admin oversight
Views	Razor views for main PAS surface and admin command center
Data	ApplicationDbContext with SQLite and database seeding
Services	StudentIdentityService for display and logging

3. Implementation Evidence

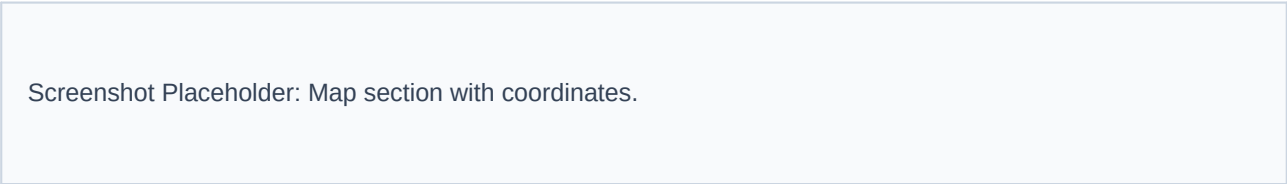
3.1 Client-Server Interaction

The home route responds with a premium PAS welcome page. The controller logs the student's full name, student ID, and Colombo, Sri Lanka for each request.



3.2 Geospatial Display

The map section displays a simplified academic department office map with department and simulated researcher coordinates.



3.3 MVC Supervisor Rendering

Supervisor records are passed from the controller to a Razor view and rendered in a premium supervisor registry table.

Screenshot Placeholder: Supervisor directory table.

3.4 Database Persistence and Search

SQLite stores seeded supervisor records and valid proposal submissions. Search supports Name and OfficeLocation queries.

Screenshot Placeholder: Successful search and no-result search.

3.5 Regex Validation

The proposal form validates StudentName and Email on the server using Data Annotation RegularExpression attributes.

Screenshot Placeholder: Invalid and valid proposal submissions.

3.6 Admin Panel

The /Admin route provides supervisor oversight, proposal intake review, module coverage, and final evidence readiness.

Screenshot Placeholder: Admin command center.

4. QA Test Matrix

ID	Area	Expected Evidence
TC-01	Welcome	PAS welcome page visible
TC-02	Logging	Identity details printed in server console
TC-03	Dashboard	Operational dashboard visible
TC-04	Map	Department map and coordinates visible
TC-05	MVC	Supervisor table rendered
TC-06	Search success	Matching records returned
TC-07	Search empty	No-results message displayed
TC-08	Invalid Regex	Validation errors displayed

ID	Area	Expected Evidence
TC-09	Valid proposal	Thank-you message displayed
TC-10	Admin	Submitted proposal visible in /Admin

5. Final Submission Checklist

Before submission, update appsettings.json with the real full name and student ID. Run dotnet restore, dotnet build, dotnet dev-certs https --trust, and dotnet run. Capture all screenshots listed in FINAL_SUBMISSION_CHECKLIST.md.

6. Conclusion

The delivered system exceeds the base brief by combining all required development and QA elements with a premium user interface, admin oversight, database-backed proposal review, and a structured evidence pack for the final PDF submission.